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A Review on Machine/Deep Learning Techniques Applied to Building Energy Simulation, Optimization and Management

Francesca Villano · G. Mauro · Alessia Pedace · Engineering, Environmental Science · Thermo · 6 March 2024

TLDR The main aim is to group the most frequent ML/DL **techniques** used in the field of building energy performance, highlighting the potentiality and limitations of each one, both fundamental aspects for future studies. Expand

45 [PDF] PDF Save Cite

A scientometric analysis and comprehensive review of artificial intelligence-based approaches for banana leaf disease detection and management

Harshit Singhal · V. Chawla · D. Tayal · S. Reddy

Agricultural and Food Sciences, Computer Science · Healthcare Engineering · 2026

TLDR A detailed overview of the global research studies is provided, highlighting key research gaps that need to be addressed and outlines future directions for building more reliable, interpretable, and comprehensive decision-support pipelines, which will guide the future research work. Expand

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Wildfire Probability Mapping in Southeastern Europe Using Deep Learning and Machine Learning Models Based on Open Satellite Data

Uroš Durlević · Velibor Ilić · Bojana Aleksova · Environmental Science, Computer Science · Applied Informatics · 9 January 2026

TLDR The results revealed that areas of very high susceptibility were most prevalent in Greece, while the smallest susceptibility percentage was recorded in Slovenia, and it was determined that the most influential factors were global horizontal irradiation, elevation, and distance from settlements. Expand

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Artificial Intelligence Techniques in Crop Yield Estimation Based on Sentinel-2 Data: A Comprehensive Survey

Muhammet Fatih Aslan · K. Sabancı · Busra Aslan

Computer Science, Agricultural and Food Sciences · Sustainability · 23 September 2024

TLDR This **review** explores the integration of **Artificial Intelligence** (AI) with Sentinel-2 satellite data in the context of precision agriculture, specifically for crop yield estimation, and identifies a notable gap in the standardization of methodologies. Expand

48 Publisher Save Cite

A Comprehensive review of Short-Term Weather Forecasting Using Artificial Intelligence Techniques Based on Internet of Things

Sulis Setiowati · Jefri Al Bukhori · Tegar Tri Hartono · Yurixa Sakhinatul Putri · R. N. Wardhani · Nuralam

Environmental Science, Computer Science · 2025 8th International Conference of Computer and… · 11 September 2025

TLDR This study concludes by proposing future directions aimed at developing lightweight, adaptive, and interpretable AI models suitable for scalable, real-time IoT deployment—thus

advancing intelligent and resilient weather forecasting systems. Expand

IEEE Save Cite

Predicting the Progression of Chronic Kidney Disease: A Systematic Review of Artificial Intelligence and Machine Learning Approaches

Fizza Khalid Lara Alsadoun +8 authors S. F. H. Bokhari Medicine, Computer Science · Cureus · 1 May 2024

TLDR This systematic **review** comprehensively evaluates the application of **artificial intelligence** (AI) and **machine learning** (ML) **techniques** for predicting CKD progression to end-stage renal disease (ESRD) or the need for renal replacement therapy. Expand

17 [PDF] PDF Save Cite

Empowering Non-Terrestrial Networks With Artificial Intelligence: A Survey

Amjad Iqbal Mau-Luen Tham +4 authors T. Dagiuklas Computer Science, Engineering · IEEE Access · 2023

TLDR This **survey** paper presents the use of **artificial intelligence** (AI) **techniques** in **learning** and adapting to the complex NTN environments, and discusses the challenges and opportunities in AI-powered NTN for 6G networks. Expand

58 PDF IEEE Save Cite

A Survey on Remote Sensing-Based Climate Change Prediction Using Artificial Intelligence Techniques

Suhana S Karthiga M Environmental Science, Computer Science · International Conference on Signals and… · 15 October 2024

TLDR There is a need for better, more powerful and easily understandable AI frameworks to handle the issue of climate change, and the reader is given an insight into the current state of AI in climate change prediction and the use of remote sensing data in this area of research. Expand

1 IEEE Save Cite

Artificial intelligence based detection and control strategies for river water pollution: A comprehensive review.

Deepa Bhatt Mahendra Swain D. Yadav Environmental Science, Computer Science · Journal of Contaminant Hydrology · 1 March 2025

TLDR This paper provides a detailed overview of **Artificial Intelligence** (AI) based **techniques** and algorithms, highlighting their practical applications in the critical domain of river water pollution diction and control and identifies emerging opportunities for future research. Expand

11 PubMed Save Cite

Resource Allocation and Load Balancing for Fog Computing Using Artificial Intelligence Techniques

Sadhna Yadav D. Verma Computer Science, Engineering · International Journal for Research in Applied… · 30 June 2025

TLDR The **survey** highlights the role of **machine learning** (ML), **deep learning** (DL), and reinforcement **learning** (RL) in intelligent resource allocation, ensuring efficient task offloading and execution and explores heuristic and nature-inspired meta-heuristic algorithms for dynamic load balancing. Expand

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